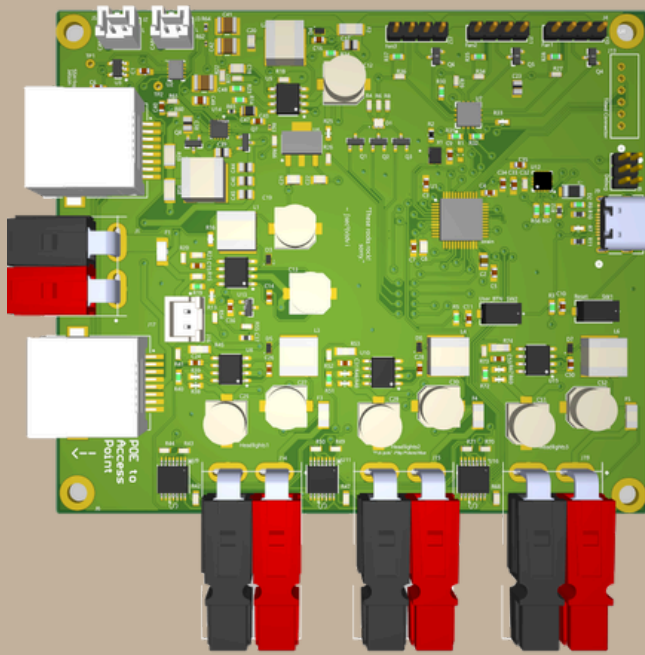


Natanel Kaplan

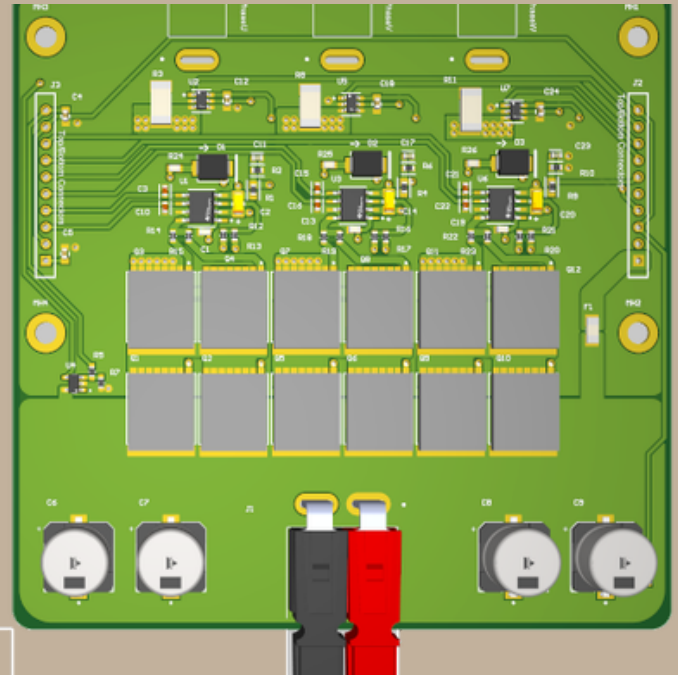
York University

Altium Designer



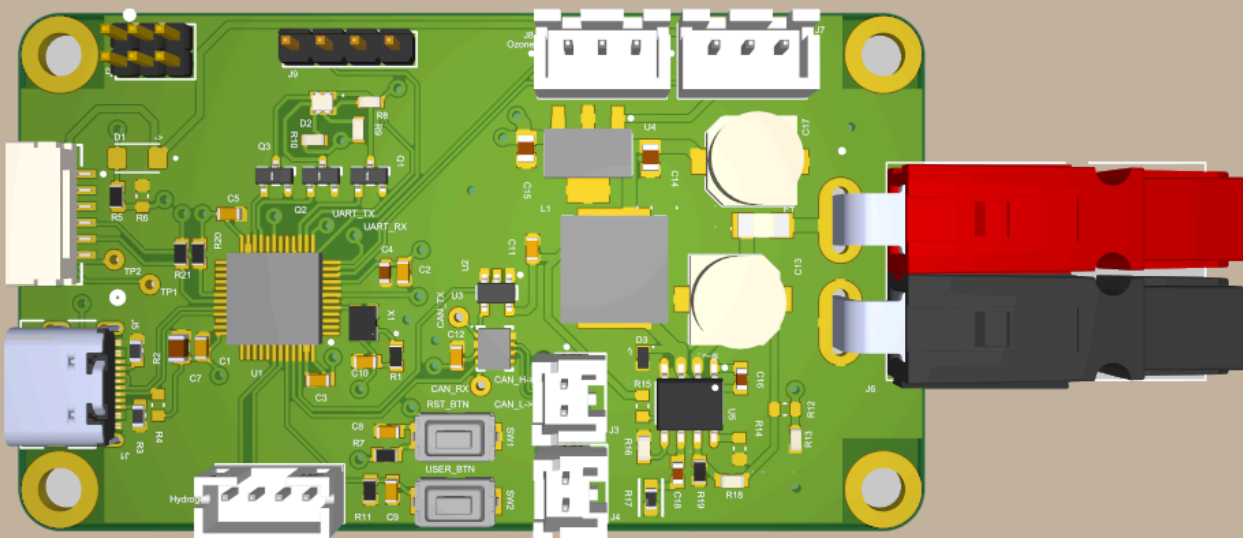
Main Control Board

STM32-based control PCB integrating a 6-axis IMU, 24V POE injector, current-controlled dimmable LEDs, dual PWM servo outputs, magnetic encoder interface, battery/USB-C voltage monitoring, and thermocouple for enclosure thermal sensing.



Brushless DC Motor Controller

High-power 3-phase BLDC motor driver sustaining 150A per phase using paralleled MOSFETs with undervoltage lockout, shoot-through protection, 2oz copper pours, and thermal vias for stable high-frequency switching under peak load.



GPS Auxiliary PCB

Compact 4-layer auxiliary PCB interfacing a NEO-M8N GPS over UART with CAN output, battery/USB-C voltage monitoring, and modular gas sensor breakouts for hydrogen and ozone detection.